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New internet speed record hits 430Tbps over fibre

Previous record broken using even less bandwidth.

By David Braue on Jan 20 2026 01:19 PM

Telecommunications researchers continue to set new speed records, with a British-Japanese team recently breaking its own records by transmitting data at 430 terabits per second (Tbps) over a single internationally standard optical fibre – while also pushing optical satellite transmission links to 2Tbps.



Researchers say they have transmitted data at 430 terabits per second (Tbps) over internationally standard optical fibre. Image: Shutterstock

One terabit, or 1 trillion bits, corresponds to about 125 gigabytes (GB), so a 430 Tbps transmission rate is equivalent to pushing nearly 54,000 gigabytes of data every second.

Pioneered by researchers at Japan's National Institute of Information and Communications Technology (NICT) and the UK's Aston University, the new technique exploits the physical characteristics of ITU-T G.654 standard optical fibre, which is widely deployed around the global internet.

Rather than transmitting data over the fibre with conventional single-mode (https://en.wikipedia.org/wiki/Single-mode_optical_fiber) techniques – which use long wavelengths in the so-called C and L transmission bands

The image shows the cover of the 'ACS Digital Pulse 2026' report. It features the ACS logo at the top left. The main headline reads 'For the human in the loop.' Below this, a sub-headline asks 'How do workers thrive in the age of AI?'. At the bottom, there is a red button that says 'Download now →'. To the right of the text is a graphic of a book cover with a blue and white abstract design and the Deloitte logo.

(https://www.acs.org.au/campaign/what-on-at-ac.html?utm_source=ia&utm_medium=website&

(<https://www.fiberlabs.com/glossary/optical-communication-band/>), – the researchers used shorter O-band wavelengths to transmit data using three-mode transmission over the same fibre optic cable.

The technique thrashed their previous record of 402Tbps, which the NICT team achieved in mid-2024.

Yet it was not only faster, but used nearly 20 per cent less of the cable's total bandwidth, the researchers said (<https://www.nict.go.jp/en/press/2025/11/11-1.html>), suggesting there is still headroom to push even more data down the cable in the future.

The new technology “extends the capacity of standard-compliant cutoff-shifted optical fibres well beyond the original design,” Dr Aleksandr Donodin, a professor at the Aston Institute of Photonic Technologies (AIPT), said in [announcing](https://www.aston.ac.uk/latest-news/aston-university-researchers-part-team-breaks-data-transmission-record-again) (<https://www.aston.ac.uk/latest-news/aston-university-researchers-part-team-breaks-data-transmission-record-again>), the latest speed record.

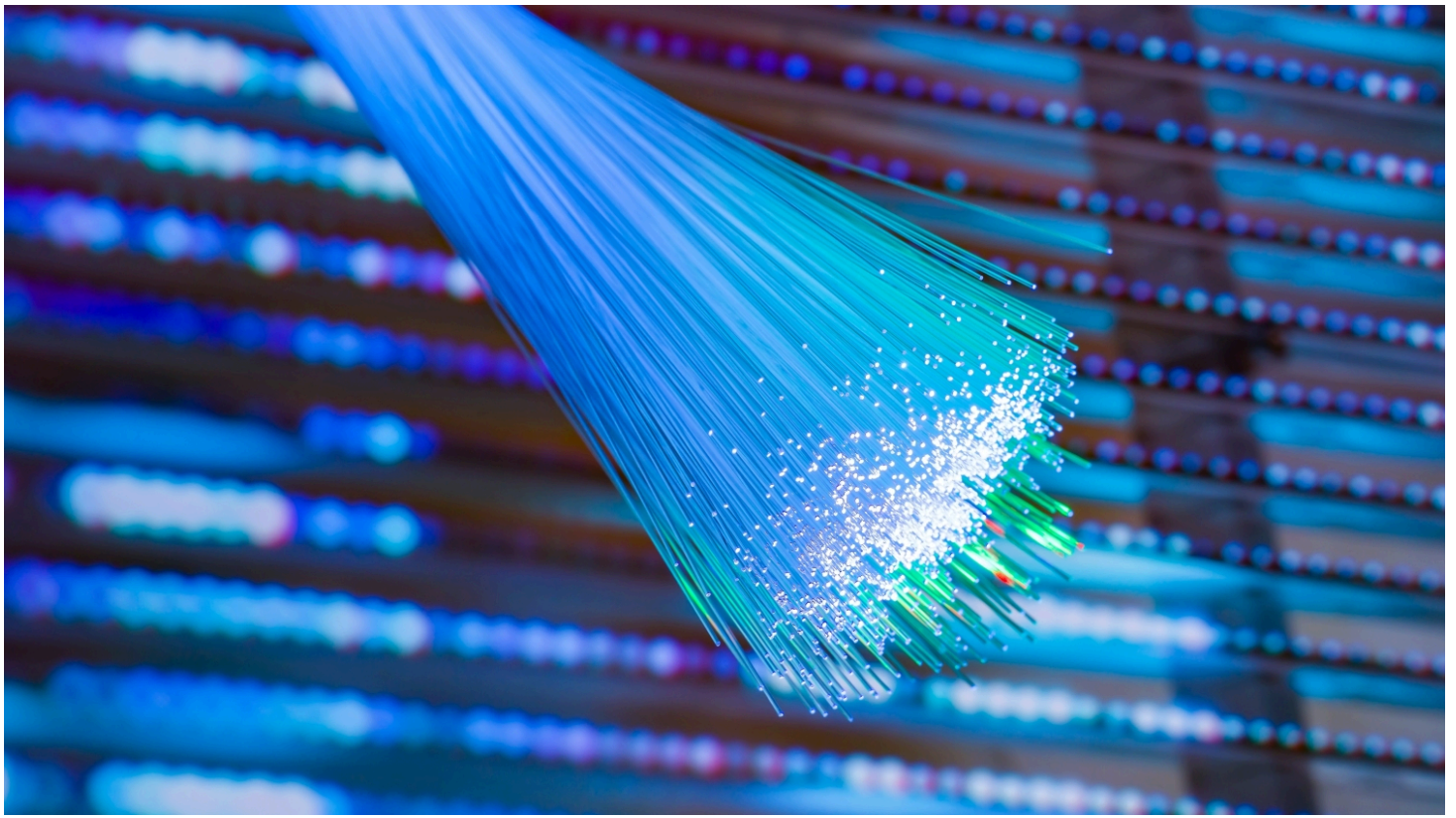
Use of O-band wavelengths “leads to a significant increase of spectral efficiency,” he said, adding that “compared to our earlier 402Tbps work, this new approach shows that we don't always need more and more spectrum to increase capacity – we can improve how efficiently we use it.”

NICT researchers have also previously broken transmission speed records using cutting-edge optical fibre technologies not yet widely used commercially, with speeds of more than 22 petabits per second (more than 22,000 terabits per second).

Bulking up global communication backbones

Faced with soaring demand for data centre connections as exploding use of data-intensive generative AI (genAI) services drives demand, global telecommunications network operators are rushing to roll out new optical fibre backbones to keep up.

So-called [400G backbones](https://www.hpe.com/emea_europe/en/what-is/400g.html) (https://www.hpe.com/emea_europe/en/what-is/400g.html), are now mainstream but rollouts of faster 800G and 1.6T networks are rapidly ramping up this year, Nokia [predicts](https://www.fierce-network.com/broadband/optical-transmission-vendors-predict-high-demand-400g-800g-2026) (<https://www.fierce-network.com/broadband/optical-transmission-vendors-predict-high-demand-400g-800g-2026>), while traffic through [subsea cables](https://www.nokia.com/blog/the-unifying-role-of-subsea-fiber-networks/) (<https://www.nokia.com/blog/the-unifying-role-of-subsea-fiber-networks/>) is increasing 30 per cent per year as data centre buildouts increase cumulative data demand by 27 per cent per year.



Researchers say their new method is 'based on standard optical fibre technology' widely used around the world. Image: Shutterstock

Industry monitor Dell'Oro Group recently increased (<https://www.delloro.com/2026-predictions-optical-transport-market/>) its forecasts for 2026, citing 15 per cent year-on-year increases in fibre optic demand and an “optimistic” outlook for the market built on “positive momentum” created by the AI industry’s frenzied growth (<https://ia.acs.org.au/article/2025/ai-data-centre-frenzy-threatens-govt-emissions-goals.html>).

Connectivity industry group IOWN Global Forum has laid down (<https://iowngf.org/evolving-networks-at-the-speed-of-light/>) a strategic vision for optical fibre development, with the ambition of reducing networks’ power consumption by 100 times, reducing end-to-end latency by 200 times, and boosting transmission capacity by 125 times.

Noting that such goals “transcend current technological realities,” the group believes “we stand at the cusp of a milestone moment in communications history”, adding that it “is working to bridge the gap between today’s technological advancements and the aspirations for a sustainable future.”

Don’t forget to look up

Yet for all the talk about pushing more data over terrestrial and undersea fibre optic cables, researchers are also looking up, with the NIST team recently publishing the results of research that increased the speed of free-space optical (FSO) communications.

Rather than relying on copper wire or optical fibre to carry signals, FSO systems use lasers (<https://ia.acs.org.au/article/2019/australia-spearheading-laser-communications.html>) to transmit data directly between two points, such as linking ground networks with satellite services like SpaceX's Starlink (<https://ia.acs.org.au/article/2024/elon-musk-s-starlink-whipping-the-competition.html>) and Amazon's Leo network (<https://www.aboutamazon.com/news/innovation-at-amazon/project-kuiper-satellite-rocket-launch-progress-updates>) – set for launch in February and favoured by Australia's NBN Co (<https://ia.acs.org.au/article/2025/nbn-co-to-replace-sky-muster-with-amazon-satellites.html>).

The NIST team demonstrated (<https://www.nict.go.jp/en/press/2025/12/16-1.html>) FSO data transmission at 2Tbps running between its Koganei, Tokyo headquarters and a small transponder 7.4 kilometres away in Chofu – highlighting not only the speed and stability of the connection, but its success overcoming environmental interference.

That interference – caused by smoke, water, and other airborne particles that affect the laser signal – remains a working challenge for FSO researchers, who this year hope to demonstrate a 10Gbps connection between a ground station and low earth orbit satellite 600km up.



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David Braue is an award-winning technology journalist who has covered Australia's technology industry since 1995. A lifelong technophile, he has written and edited content for a broad range of audiences across myriad consumer and business topics, with a particular focus on managing the intersection of technological innovation and business transformation. He has twice won Best IT Journalist at the Australian IT Journalism awards, and was named Best Technology Journalist at the 2024 Australian Technologies Competition.

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